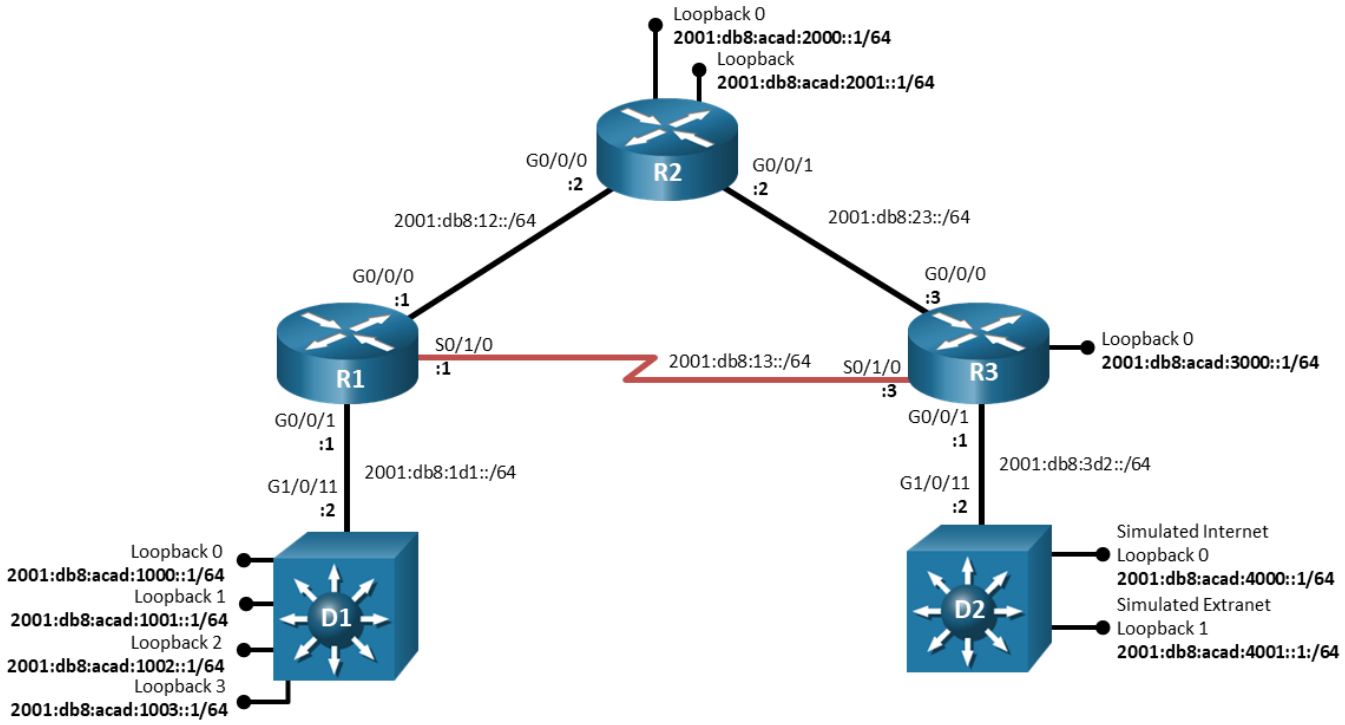
Lab - Troubleshoot EIGRP for IPv6

Topology



Addressing Table

Device	Interface	IPv6 Address/Prefix Length	Link-Local Address
R1	G0/0/0	2001:db8:12::1/64	fe80::1:1
	G0/0/1	2001:db8:1d1::1/64	fe80::1:2
	S0/1/0	2001:db8:13::1/64	fe80::1:3
R2	G0/0/0	2001:db8:12::2/64	fe80::2:1
	G0/0/1	2001:db8:23::2/64	fe80::2:2
	Loopback 0	2001:db8:acad:2000::1/64	fe80::2:3
	Loopback 1	2001:db8:acad:2001::1/64	fe80::2:4
R3	G0/0/0	2001:db8:23::3/64	fe80::3:1
	G0/0/1	2001:db8:3d2::1/64	fe80::3:2
	S0/1/0	2001:db8:13::3/64	fe80::3:3
	Loopback 0	2001:db8:acad:3000::1/64	fe80::3:4
D1	G1/0/11	2001:db8:1d1::2/64	fe80::d1:1

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Device	Interface	IPv6 Address/Prefix Length	Link-Local Address
	Loopback 0	2001:db8:acad:1000::1/64	fe80::d1:2
	Loopback 1	2001:db8:acad:1001::1/64	fe80::d1:3
	Loopback 2	2001:db8:acad:1002::1/64	fe80::d1:4
	Loopback 3	2001:db8:acad:1003::1/64	fe80::d1:5
D2	G1/0/11	2001:db8:3d2::2/64	fe80::d2:1
	Loopback 0	2001:db8:acad:4000::1/64	fe80::d2:2
	Loopback 1	2001:db8:acad:4001::1/64	fe80::d2:3

Objectives

Troubleshoot network issues related to the configuration and operation of EIGRP for IPv6.

Background / Scenario

A mid-size enterprise is in the middle of standardizing its IPv6 IGP across the campus and branches using EIGRP for IPv6. The design calls for aggressive route summarization to keep the internal routing tables small, while ensuring only a default route to the internet is learned internally. The partner extranet connected at D2 must remain hidden from the rest of the network, as must a small, isolated enclave behind R3. The large branch behind D1 is growing, so the plan is to advertise a single summary that covers both existing subnets and several future additions, avoiding routing table churn when those networks come online. R1, R2, R3, D1, and D2 form the EIGRP neighbor core that should reflect this design.

Your tasks are to FIND the error(s), document your findings and the command(s) or method(s) used to fix them, FIX the issue(s) presented here and then test the network to ensure both of the following conditions are met:

1) the complaint received in the ticket is resolved
2) full reachability is restored

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.5 (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.5 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.5c universal image or comparable)
- 2 Switches (Cisco 9300L with Cisco IOS XE Release 17.12.5 universal image or comparable)
- 1 PC (Choice of operating system with terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

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Instructions

Part 1: Trouble Ticket 4.1.2.1

Scenario:

During a maintenance window, a junior administrator attempted to implement the summarization and route-filtering plan but misapplied several settings. The morning after, operations reported that some sites could not reach the internet via D2, internal routers started seeing partner routes they weren't supposed to, and the routing tables were larger than expected. Additionally, planned expansion networks at the large branch were not covered by a working summary as intended. Your task is to audit and correct the EIGRP for IPv6 configuration so that D1 properly advertises its branch summary, R2 advertises the correct summary for its two loopback networks to both sides, R3 keeps its enclave off the IGP, and D2 redistributes only the default route while keeping the extranet hidden. Once fixed, the internet path via D2 must work, the enclave and extranet must remain invisible, and the routing table should reflect the intended summaries, including future branch networks.

Be sure you are using devices that have been connected and prepared by your instructor. Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/4.1.2.1-r1-config.txt running-config`
R2	`copy flash:/enarsi/4.1.2.1-r2-config.txt running-config`
R3	`copy flash:/enarsi/4.1.2.1-r3-config.txt running-config`
D1	`copy flash:/enarsi/4.1.2.1-d1-config.txt running-config`
D2	`copy flash:/enarsi/4.1.2.1-d2-config.txt running-config`

- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- When you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the **reset.now** privileged EXEC command. This script will clear your configurations and reload the devices.

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